◆ ANALYTICS PORTFOLIO

RETAIL ANALYTICS · TABLEAU CASE STUDY

Retail Profit Leakage Analysis

A profitability review of 15,000 retail transactions to locate where margin is strongest, where losses occur, and what to test next.

Data period · Jan 2024–Dec 2025 | 15,000 transactions | Tableau dashboard + action plan

NET SALES	TOTAL PROFIT
₹61.56M	₹20.31M
₹61,561,351.66	₹20,311,566.15

PROFIT MARGIN	LOSS TRANSACTIONS
32.99%	42
Profit ÷ net sales	Profit below zero

01 · PERFORMANCE

Where profit is being made

The category and discount views help separate high sales from healthy unit economics. Electronics is the clearest area to investigate.

Profit margin by category

Category margin varies from 22.70% to 45.52%.

Clothing	**45.52%**
Sports	**40.99%**
Home & Kitchen	**37.34%**
Office Supplies	**29.45%**
Electronics	**22.70%**

Electronics net sales **₹28.33M · 46.0% of total**

Margin by discount quartile

Equal-sized groups of 3,750 transactions, ordered by discount.

Q1 · 2.57% off	**39.30%**
Q2 · 8.99% off	**35.48%**
Q3 · 13.95% off	**31.12%**
Q4 · 23.15% off	**23.67%**

> Q4 margin is 15.63 percentage points below Q1. This is a historical association, not proof that discounts caused lower margin.

42

Loss-making transactions

All occurred in Electronics. Investigate order-level discount, cost, quantity, and sales channel before changing the product range.

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Mouse orders to review first

Mouse had the most loss transactions, followed by Headphones (10), Keyboard (8), Bluetooth Speaker (8), and Monitor (3).

₹3.05M

Q4 still made a profit

The highest-discount quartile remained profitable overall. The priority is testing better offer economics, not assuming every discount loses money.

Category performance

Sales, profit, and margin together show the trade-offs behind the headline totals.

CATEGORY	NET SALES	PROFIT	MARGIN
Clothing	₹16.16M	₹7.36M	45.52%
Electronics	₹28.33M	₹6.43M	22.70%
Home & Kitchen	₹7.76M	₹2.90M	37.34%
Office Supplies	₹1.65M	₹0.49M	29.45%
Sports	₹7.66M	₹3.14M	40.99%

Dashboard views

The Tableau workbook brings together the core category, product, time, and KPI views:

- Revenue vs Profit by Category
- Profit Margin by Category
- Monthly Revenue Trend
- Discount by Product
- Four KPI sheets

02 · FROM INSIGHT TO ACTION

A focused 30-day plan

Start with the 42 loss orders, then run a controlled discount test and add a weekly exception review.

01

Classify the 42 loss orders

Review product, discount, region, channel, quantity, and unit cost. Tag each loss with a reason such as promotion, price override, cost mismatch, or data issue.

WEEKS 1–2 · PRICING + FINANCE

02

Test Electronics offer guardrails

Compare a lower discount tier or approval step for deeper promotions with a matched baseline. Track profit per order and conversion together.

WEEKS 2–4 · COMMERCIAL + PRICING

03

Monitor order-level exceptions

Review negative-profit orders and discount exceptions weekly, with product, region, and channel drilldowns. Track loss count and total negative profit.

WEEKLY · BI + FINANCE

04

Protect total business performance

Review category sales, profit, margin, and discount quartiles together. Set discount thresholds only after confirming cost and promotion economics.

MONTHLY · CATEGORY LEADS

How to interpret the analysis

These are descriptive summaries from the existing retail sales data. Product groups remain profitable in aggregate even when some individual transactions lose money. Discount guardrails are a test to validate, not a guaranteed outcome.

03 · WORK PRODUCTS

Explore the project files

The dashboard and executive action plan are available alongside this portfolio page.

Tableau dashboard workbook
Retail Profit Leakage Analysis · TWB ↗

Executive action plan
Verified findings and 30-day recommendations · DOCX ↗

The Tableau workbook connects to the existing local MySQL data source; the source rows are not embedded in this file.

Retail Profit Leakage Analysis · Portfolio case study

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